MABRURI ACHYARUL

Semiconductor Device Researcher · RRAM & Neuromorphic Computing · Youth Leader & Mentor

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PROFILE

Graduate research student at Korea University's Advanced Semiconductor Laboratory on the SK Global Scholarship, working on resistive memory (RRAM) devices for energy-efficient, brain-inspired computing. Experienced across the full chain from device fabrication and measurement to modeling and hardware-aware neural networks. Fully funded study and research in India, Russia, Korea and Singapore, and active in youth development in Indonesia since 2018, leading an NGO that has mentored 100+ young people.

EDUCATION

Korea University, Seoul, Republic of Korea 2025 – present

M.S. (Research), Semiconductor and Nano Systems, School of Electrical Engineering

- **SK Global Scholarship** (Korea University / SK hynix), full scholarship. Advanced Semiconductor Laboratory under Prof. Tae Geun Kim.

Delhi Technological University, New Delhi, India 2021 – 2025

B.Tech., Engineering Physics. First Division with Distinction

- **ICCR Scholarship** (Government of India), full scholarship. Thesis: "Temperature Dependence of Complete Photonic Band Gap in 2D Photonic Crystal Structures."

RESEARCH EXPERIENCE

Graduate Researcher, Advanced Semiconductor Laboratory, Korea University 2025 – present

- Fabricate and characterize lead-free metal-halide RRAM devices by sputtering and Keithley 4200-SCS I–V and pulse measurements; analyze switching, variability and conduction mechanisms.
- Member of the lab's PRAM (phase-change memory) team: device-to-system simulation of the group's phase-change heterostructure (PCH) memories.
- Investigate selector concepts for RRAM crossbar arrays to suppress sneak-path currents; simulate devices and arrays in LTspice, Python and COMSOL Multiphysics.

Research Fellow, InteRussia Engineering Fellowship, Novosibirsk State University, Russia Jun 2025

- Selected fellow in Modern Quantum and Information Technologies in Electronics and Photonics; built a Python simulation project with an international cohort.

Researcher, Advanced Simulation Lab, TIFAC-CORE, Delhi Technological University 2024 – 2025

- Modeled temperature-dependent photonic band gaps in Si/GaAs photonic crystals for sensing (MPB, MEEP); published in the Journal of Modern Optics.

Research Intern, Silicon Photonics Research Group, LNMIIT Jaipur, India May – Jul 2024

- Modeled carrier mobility and the indirect-to-direct band-gap crossover in $Si_{1-x}Sn_x$ alloys in MATLAB.

KEY RESEARCH PROJECTS

Metal-halide RRAM: from measured device to crossbar and neural network 2026

- Fitted conduction and weight-update models to lab CsX RRAM data; simulated passive crossbars with wire resistance and ADCs; a binary CNN kept 91.5% CIFAR-10 accuracy on device states.

Phase-change heterostructure (PCH) memory for in-memory computing 2026

- Built a device-to-chip pipeline for the group's multi-level PCH devices ($NiTe_2$/$MoTe_2$, MoS_2, $HfTe_2$, $TiTe_2$) vs. a GST baseline: crossbar MAC with drift and IR drop, device-aware VGG8 on CIFAR-10, and NeuroSim TOPS/W.

Drift compensation for phase-change-memory analog inference 2026

- Simulated one year of PCM drift in VGG8 inference with IBM aihwkit; global drift compensation plus adaptive batch-norm restored accuracy from 53% to 90%.

NEUROM: memristive reservoir-computing benchmark across six device models 2025 – 2026

- Echo-state reservoirs with STDP initialization under Monte Carlo device variability; tasks: sequential MNIST, Mackey–Glass forecasting, spoken digits.

Memristor compact-model framework and reliability study 2025 – 2026

- Python library of 21 compact models (window-function, Yakopcic, TEAM, VTEAM, Knowm); identified and fixed three hidden failure modes that silently corrupt LTP/LTD simulations.

Spiking neural network on measured memristor conductance levels 2026

- Trained a leaky integrate-and-fire network in snnTorch (96.1% on MNIST) and tested it on three real devices' conductance levels.

Self-rectifying memristor crossbar pipeline 2026

- Replicated Sharma et al. (Advanced Materials, 2026) and applied it to three measured devices (readout margin, array size, CNN accuracy).

Physical reservoir computing with Ag/MoS_2/Au CBRAM (LTspice + Python) 2026

- Replicated the Tanaka & Nakane (2022) reservoir; compared discrete-switching CBRAM with a continuous Yakopcic model.

Machine-learning surrogate for STDP in drift–diffusive memristor synapses 2026

- Simulated 2,875 circuit configurations; XGBoost models (99.2% accuracy) map STDP behavior without re-simulation.

Reproductions and tools 2025 – 2026

- Reproduced Vourkas (2025) netlists NL1–NL5, the Scientific Reports (2026) RRAM compact model and a lead-free NKN/HfO_2 STDP model; β-VAE reproduction as a course project (with J. Oh).
- Extended the open-source PyNeuroMemSim (LTspice–Python) simulator to four memristor models; ported Keithley 4200-SCS scripts from C to Python; reproduced ML prediction of SET voltage in honey-based RRAM.

LEADERSHIP & PROFESSIONAL EXPERIENCE

Managing Director, Gerakan Tunas Bangsa Indonesia (youth-development NGO) Dec 2021 – Aug 2024

- Led and managed the NGO's day-to-day operations, programs and volunteer team; formulated and set its policies and core values.
- Coordinated internal and external stakeholders to keep the movement active, empowering and sustainable.
- Supervised mentoring, coaching and scholarship-awareness activities reaching 100+ young people.

Human Quality & Development Lead, Gerakan Tunas Bangsa Indonesia 2018 – 2021

- Standardized member-development quality and created a system to evaluate each mentee's progress.
- Launched new programs (English classes, coaching, sharing sessions, webinars and workshops) for 100+ mentees.

Speaker & Mentor: Scholarships, Higher Education and Personal Development 2021 – present

- Invited speaker, e.g. “Unlock Your Dreams Through Education” (2026); “Developing a Successful Mindset Towards a Dream” (2025); ICCR scholarship mentoring for ~100 participants with PPI India.

Board of Supervisors, Indonesian Students Association in India (PPI India) Jul 2024 – Jun 2025

- Supervised the association's programs and governance for Indonesian students across India.

Asia Region Representative, DTU International Students Society Nov 2022 – Aug 2024

- Represented international students from the Asia region in the society's Chamber of Representatives.

INTERNATIONAL PROGRAMS & FELLOWSHIPS

- **International Festival of Youth (IFY) 2026**, Yekaterinburg, Russia: Indonesian delegate Sep 2026
- **SEA–Europe Youth Partnership for the Goals 2026**, Singapore (ASEF, KU Leuven) 2026
- **Honorary Young Diplomat**, Jeju International Future Generation Forum (UNITAR), Korea Nov 2025
- **Aspire Leaders Program**, Cohort 1 2025
- **Kectil Leadership Program** (Kectil Colleague) and **UPG Leadership Program** (Leadership Candidate) 2023

HONORS & AWARDS

- **SK Global Scholarship**, Korea University / SK hynix (full scholarship) 2025 – present
- **ICCR Scholarship**, Indian Council for Cultural Relations, Government of India (full scholarship) 2021 – 2025
- **Top 6 Nominee**, Youth Change Maker Award; **Top 27**, Paragon Innovation Awards 2021
- **Best Candidate**, LeadID (Leaders Academy Indonesia) by pemimpin.id 2021

PUBLICATIONS & PRESENTATIONS

- A. Mabruri et al., “Investigation of Temperature Dependence of Complete Photonic Band Gap in 2D Photonic Crystal with Sensing Application,” *Journal of Modern Optics*, 2025. DOI: 10.1080/09500340.2025.2542218
- Poster presentation, 3rd International Conference on Advanced Functional Materials and Devices for Sustainable Development (AFMD), 2025.

SKILLS & CERTIFICATIONS

Lab & devices: sputtering, photolithography, thermal evaporator, RRAM fabrication, Keithley 4200-SCS I–V, pulse testing

Simulation & AI: Python (PyTorch, snnTorch, scikit-learn), MATLAB, LTspice, COMSOL Multiphysics, MPB/MEEP; neural networks and data analysis

Certifications: Quantum Machine Learning, Hasso Plattner Institute (2023); Molecules and Materials for the Energy of Tomorrow, Université Paris-Saclay (2023); QSilver35 Quantum Computing & Programming, QWorld (2026)

Languages: Javanese and Indonesian (native), English (fluent), Korean (beginner)